

Rotations, orbits, and shuffles

Week 3 allowed arbitrary permutations. Week 4 fixes a circular order and studies shifts: what controls their cycles and return times? The extra changes addition to multiplication through a real card shuffle.

Materials and preparation

Print 3 K-1, 3 middle, 1 upper, 1 extra reserve, and this guide. Bring seven counters, pencils, scratch paper, and cards 0-7 (another eight for the sixteen-card extension). For the K six-picture extension, draw shapes on scraps. Hop counters on the printed fixed rings; no movable wheel or brad is needed. Draw strips on scratch paper for the optional proof.

Readiness

K-1: match shapes, count hops to three with help; no independent reading. Reason about repeated movement and impossible landings. **2-3:** A-J familiarity, count up to ten; read-aloud removes word-reading barriers. Explain disjoint routes and use a common key. **4-5:** count to eighteen for finite rings. The optional general proof additionally needs division/gcd and the adult-supported coprime lemma below. Distinguish an orbit from the whole permutation. **Extra:** track numbered positions and remainders; powers can be explained as repeated doubling. The core eight-card proof needs only 2,4,8 and remainder 1.

A shared 60-minute hour

0-10: play with counters and rings, trying hop lengths. **10-15:** define clockwise and count movements, not the starting place. Launch: “Can a step rule miss some places forever?” **15-35:** K compares one/two-step routes; middle tries the code clue; upper explores the 12-ring. Adult A anchors K; organizer visits middle and upper. **35-40:** movement/reset: stand, stretch, leave work ready to return to. **40-55:** continue orbit comparisons or an optional proof below; offer the shuffle as an alternative. **55-60:** share a route, obstruction, or explanation; tidy counters and cards. The twelve-row table is optional: inverse shifts share return times. Move to explanation when ready. An adult should discuss the upper child’s conjecture; the three pages are choices, not a timed test.

Hints and common confusions

Count spaces moved, not the starting label. Stop an orbit when its start returns; then choose an unused start. Ask how many spaces and whole laps were traveled. In shuffling, distinguish a card label from its current position.

Satisfying stops / optional proof

K-1 stop: compare one/two-hop routes and undo the hidden turn. **Optional:** explain missed shapes using separate loops.

2-3 stop: decode and compare +2/+3 routes. **Optional:** prove with alternate markings that +2 misses half, yet has an inverse.

4-5 stop: explain +4/+8 returns and loops on twelve places. **Optional:** prove the gcd formula with the adult strip scaffold; otherwise label the general rule a conjecture or supplied theorem.

Exact routes and the gcd rule

K-1

From circle, one hop visits triangle, square, star, circle (four turns); two hops alternate circle/square (two turns), never landing on triangle or star. A secret circle-to-square shift is two hops, so triangle becomes star, square becomes circle, and star becomes triangle. Two hops undo themselves. Three-hop turns follow circle, star, square, triangle, circle, visiting all four. On six positions, among steps 1 through 5, only 1 and 5 visit all six; 2 and 4 give two loops of length 3; 3 gives three loops of length 2.

Middle

A to D is +3 on A–J. GDG decodes to **DAD**; BAG encodes to **EDJ**. B becomes E, so the additional B-to-F clue contradicts the fixed-shift rule. Repeated +2 gives A,C,E,G,I,A and B,D,F,H,J,B. Marking alternate letters explains why the two routes cannot meet: adding two preserves that marking. Repeated +3 gives A,D,G,J,C,F,I,B,E,H,A. The steps visiting all ten are **1,3,7,9**. Any shift is reversible: +2 is undone by -2, or +8. Skipping some letters along one repeated orbit is not information loss.

Upper: key and complete classification

9 to 1 gives +4, which takes 2 to 6. Outputs 1,4,10 decode to **9,0,6**. Its four loops are (0,4,8), (1,5,9), (2,6,10), (3,7,11), each returning in three turns.

Shift k	0	1	2	3	4	5	6	7	8	9	10	11
Return time	1	12	6	4	3	12	2	12	3	4	6	12
Number of loops	12	1	2	3	4	1	6	1	4	3	2	1

Steps 1,5,7,11 visit all twelve. The zero rule is the identity and has first *positive* return time 1, not zero. For eighteen positions, +6 returns after 3; +5 after 18.

Why the formula works

For +8 on 12 places, return means $3 \mid 2t$. Draw three t-strips, totaling $3t$. If the first two strips total a multiple of 3, removing them leaves t a multiple of 3. Thus the first positive return is 3. This proves this case for every t . The optional general bridge is on page 4; do not treat a completed table as its proof.

From turning to shuffling

Upper: inverse and composition

On twelve places, +4 is undone by +8 (or -4). For every starting x , +4 followed by +7 gives $x+11$ modulo 12; reversing the order gives the same because integer addition commutes. Every shift is bijective, even if its orbit from 0 is a proper subset. The visit-everywhere condition is $\gcd(n, k) = 1$, not a condition for reversibility. This distinction is central: a substitution permutation can have several cycles.

Extra: eight cards, three shuffles

Rows, left to right with the leftmost card the top:

Shuffle	Row
0	0 1 2 3 4 5 6 7
1	0 4 1 5 2 6 3 7
2	0 2 4 6 1 3 5 7
3	0 1 2 3 4 5 6 7

The old-to-new position map is **0,2,4,6,1,3,5,7**. Its cycles are fixed 0, fixed 7, (1,2,4), and (3,6,5). Thus the whole shuffle returns after 3 and not earlier. This is also a direct week 3 loop argument. For old position x in the first half (0–3), the new position is $2x$. In the second half (4–7), it is $2(x - 4) + 1 = 2x - 7$. For x below 7 these are exactly $2x \bmod 7$. Position 7 stays 7, separately; it must not be merged with residue 0. Repetition gives $2^t x \bmod 7$. The powers 2,4,8 have remainders 2,4,1, so the first identity occurs at $t=3$. Tracking position 1 rules out earlier global returns, and remainder 1 fixes all positions 0–6.

Extra predictions

Six cards use modulus 5. Powers 2,4,8,16 give remainders 2,4,3,1, so the order is **4**. Sixteen cards use modulus 15: 2,4,8,16 give 2,4,8,1, again order **4**. In general an out-shuffle on $2m$ cards maps positions below $2m - 1$ by doubling modulo $2m - 1$, fixes the final position, and has the multiplicative order of 2 as return time (the two-card identity is a trivial edge case). More cards can therefore have the same or even a smaller return time.

A useful additional contrast

Random shuffling asks about a distribution over possible orders. This perfect shuffle is one precisely defined permutation repeated deterministically. Its return time says nothing by itself about how mixed a random shuffle is. Do not introduce chance unless it becomes a separate, explicitly changed rule.

Mathematical lineage and reuse

Optional adult bridge: coprime divisibility

If positive a, b are coprime and $a \mid bt$, compare base strips a, b with scaled strips at, bt . Repeatedly subtract the smaller base strip from the larger, doing the same to the scaled strips. Both scaled lengths stay multiples of a . The positive base sum decreases, so the process ends with equal lengths c . Common divisors are unchanged: a divisor of x, y also divides $x-y, y$, and conversely by adding. Thus $c=1$, since the original lengths were coprime. The scaled process ends at t , which is therefore a multiple of a . Conversely, $a \mid t$ plainly implies $a \mid bt$.

Try 5,3 first: $5-3=2$, then $3-2=1$. On scaled strips this leaves $3t - (5t - 3t) = t$. Equivalently, $2 \cdot 3t - 5t = t$; if $5 \mid 3t$, the remainder is a multiple of 5. Ask the child to explain each removal. This is additional readiness, not assumed knowledge of prime factors.

For $k>0$, set $d = \gcd(n, k)$, $n = da$, $k = db$. Return means $a \mid bt$; the lemma gives $a \mid t$, hence first return n/d . Every start has this length, so there are d loops. For $k=0$, every point is fixed. If only the 3-and-2 case was explained, record that proof and label the general rule conjectured or given as a theorem. Full strip prompts and the Week 9 application: `plans/coprime-divisibility-scaffold.md`.

Sources and boundaries

Judson, *Abstract Algebra: Theory and Applications*, Ch. 4, §4.1, Theorem 4.13, pp. 48–49, gives the cyclic order formula: textbook. Our rings are its additive version. **Diaconis, Graham, Kantor (1983)**, *The mathematics of perfect shuffles*, §2, Lemma 1, p. 177, proves the out-shuffle order via doubling: original paper. The extra proves this small return-time case and position rule, not the paper's shuffle-group classification. The plan has full research references. Rozhkovskaya, Lesson 2, problems 2.3–2.7, supplies the shift-cipher entry. *Math Circle by the Bay*, preface viii–x, supports manipulatives, changing pace, and reserves; Rozhkovskaya Lesson 8 records copying slowing the activity. Our rings, proof scaffolds, and hour are adaptations. Lowell Handout 6, problem 6.7, and Handout 7, problem 7.8, use a different card-ordering solitaire. Ask about recognition; do not assume this shuffle was taught.

Use history and checks

Record actual steps, ring sizes, and extensions used as F04-K/M/U/X-v2. Keep 26-letter Caesar messages as a later alternative, not extra arithmetic for early finishers. Future years can compare prime/composite ring sizes, affine maps, in-shuffles, or perfect-shuffle groups; avoid replaying the same small cases. `verify.py` checks all 12 shifts, the 10-letter routes, message examples, the 18-ring predictions, shuffle rows, position formulas, and first return times for 6, 8, 16 cards. The facilitator proofs establish why the patterns hold.